Reliable data transfer

CS2105 · Lecture 4

Demo course material, written for OpenPapr

The problem

- The network layer may corrupt, lose or reorder packets
- The transport layer must still deliver the data correctly and in order

The tools

- Checksums detect corrupted packets
- ACKs tell the sender what arrived
- Sequence numbers spot duplicates
- Timers recover lost packets

Stop-and-wait (rdt 3.0)

- Send one packet, then wait for its ACK
- Retransmit when the timer expires
- A 1-bit sequence number (0, 1) is enough

Stop-and-wait is slow

- Utilisation U = (L/R) / (RTT + L/R)
- L = 8000 bits, R = 1 Gbps, RTT = 30 ms gives U = 0.00027
- The link sits idle while the sender waits

Pipelining

- Let several unACKed packets be in flight at once
- Needs a range of sequence numbers and buffering
- Two ways to do it: Go-Back-N and Selective Repeat

Go-Back-N: the sender

- Up to N unACKed packets in the window
- Cumulative ACK: ACK n means every packet up to n arrived
- One timer, for the oldest unACKed packet
- On timeout, resend every packet in the window

Go-Back-N: the receiver

- Accepts only the next in-order packet
- Discards out-of-order packets and buffers nothing
- Re-sends the ACK for the last in-order packet

Go-Back-N in action

- Window N = 4; packet 2 is lost
- Packets 3, 4 and 5 arrive out of order and are discarded; the receiver repeats ACK 1
- Timeout: the sender resends 2, 3, 4 and 5

Selective Repeat

- The receiver ACKs and buffers each packet on its own
- The sender keeps a timer per packet and resends only the lost ones
- The window must be at most half the sequence number space

Summary

- Checksums, ACKs, sequence numbers and timers
- Pipelining fixes the utilisation of stop-and-wait
- GBN: simple receiver, more resending
- SR: less resending, more buffering